

# Band Structure Calculations of Two-dimensional Kagome Lattice by Tight Binding Model

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## I. INTRODUCTION

The kagome lattice in this sense consists of the vertices and edges of the trihexagonal tiling [1].

Different from the two type of sites in Honeycomb lattice, the Kagome lattice has 3 types of sites, *A* (blue), *B* (red), and *C* (green), respectively. The three sites in the unit cell form a regular

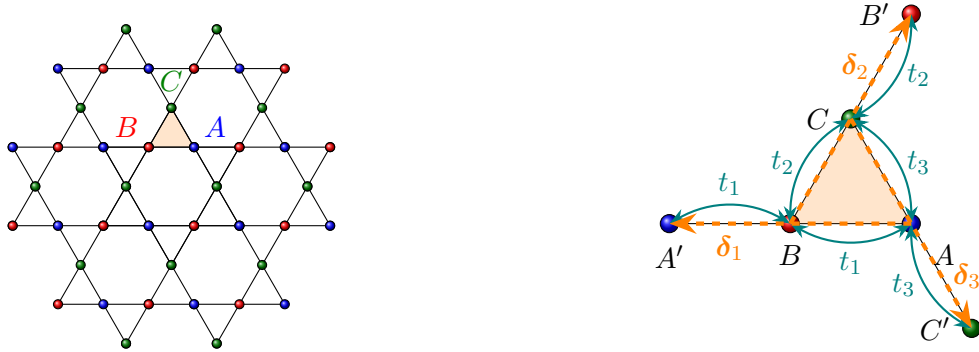


FIG. 1: Top view of the Kagome Lattice and tight-binding model with the NN hopping

triangle, which is shaded in Fig. 1. If we simplify taking the unit cell located at  $\mathbf{r}$ . Then, each site in the unit cell has 2 NN hoppings, pointing out by the lattice vectors

$$\boldsymbol{\delta}_1 = a(-2, 0), \quad \boldsymbol{\delta}_2 = a(1, \sqrt{3}), \quad \boldsymbol{\delta}_3 = a(1, -\sqrt{3}). \quad (1)$$

where  $a$  is the lattice constant, the distance between any two NN lattices.

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## II. MODEL HAMILTONIAN

In this model, consider the nearest-neighbor (NN) hopping with different hopping magnitude  $t_1$ ,  $t_2$ , and  $t_3$ . The Hamiltonian illustrates the tight-binding model in terms of second quantization

$$\mathcal{H} = - \sum_{\mathbf{r}} [t_1(a_{\mathbf{r}}^\dagger b_{\mathbf{r}} + b_{\mathbf{r}}^\dagger a_{\mathbf{r}+\delta_1}) + t_2(b_{\mathbf{r}}^\dagger c_{\mathbf{r}} + c_{\mathbf{r}}^\dagger b_{\mathbf{r}+\delta_2}) + t_3(c_{\mathbf{r}}^\dagger a_{\mathbf{r}} + a_{\mathbf{r}}^\dagger c_{\mathbf{r}+\delta_3})] + \text{H.c.} \quad (2)$$

Assume there are  $N$   $A$ -,  $B$ -, and  $C$ - type sites in total. Applying the Fourier transformation to the Fermion operators

$$\begin{aligned} \hat{a}_i &= \frac{1}{\sqrt{N}} \sum_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{r}_i} \hat{a}_{\mathbf{k}}, & \hat{b}_i &= \frac{1}{\sqrt{N}} \sum_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{r}_i} \hat{b}_{\mathbf{k}}, & \hat{c}_i &= \frac{1}{\sqrt{N}} \sum_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{r}_i} \hat{c}_{\mathbf{k}}, \\ \hat{a}_{i+\delta_1} &= \frac{1}{\sqrt{N}} \sum_{\mathbf{k}} e^{i\mathbf{k} \cdot (\mathbf{r}_i + \delta_1)} \hat{a}_{\mathbf{k}}, & \hat{b}_{i+\delta_2} &= \frac{1}{\sqrt{N}} \sum_{\mathbf{k}} e^{i\mathbf{k} \cdot (\mathbf{r}_i + \delta_2)} \hat{b}_{\mathbf{k}}, & \hat{c}_{i+\delta_3} &= \frac{1}{\sqrt{N}} \sum_{\mathbf{k}} e^{i\mathbf{k} \cdot (\mathbf{r}_i + \delta_3)} \hat{c}_{\mathbf{k}}. \end{aligned}$$

Similar for their Hermite conjugates

$$\begin{aligned} \hat{a}_i^\dagger &= \frac{1}{\sqrt{N}} \sum_{\mathbf{k}} e^{-i\mathbf{k} \cdot \mathbf{r}_i} \hat{a}_{\mathbf{k}}^\dagger, & \hat{b}_i^\dagger &= \frac{1}{\sqrt{N}} \sum_{\mathbf{k}} e^{-i\mathbf{k} \cdot \mathbf{r}_i} \hat{b}_{\mathbf{k}}^\dagger, & \hat{c}_i^\dagger &= \frac{1}{\sqrt{N}} \sum_{\mathbf{k}} e^{-i\mathbf{k} \cdot \mathbf{r}_i} \hat{c}_{\mathbf{k}}^\dagger, \\ \hat{a}_{i+\delta_1}^\dagger &= \frac{1}{\sqrt{N}} \sum_{\mathbf{k}} e^{-i\mathbf{k} \cdot (\mathbf{r}_i + \delta_1)} \hat{a}_{\mathbf{k}}^\dagger, & \hat{b}_{i+\delta_2}^\dagger &= \frac{1}{\sqrt{N}} \sum_{\mathbf{k}} e^{-i\mathbf{k} \cdot (\mathbf{r}_i + \delta_2)} \hat{b}_{\mathbf{k}}^\dagger, & \hat{c}_{i+\delta_3}^\dagger &= \frac{1}{\sqrt{N}} \sum_{\mathbf{k}} e^{-i\mathbf{k} \cdot (\mathbf{r}_i + \delta_3)} \hat{c}_{\mathbf{k}}^\dagger. \end{aligned}$$

Substituting the transformed operators into the Hamiltonian, one can obtain  $\mathcal{H} = \sum_{\mathbf{k}} \Psi_{\mathbf{k}}^\dagger \tilde{\mathcal{H}} \Psi_{\mathbf{k}}$ , where

$$\tilde{\mathcal{H}} = - \begin{pmatrix} 0 & t_1(1 + e^{ik_1}) & t_2(1 + e^{ik_2}) \\ t_1(1 + e^{-ik_1}) & 0 & t_3(1 + e^{-ik_3}) \\ t_2(1 + e^{ik_2}) & t_3(1 + e^{ik_3}) & 0 \end{pmatrix}. \quad (3)$$

## III. ADVANCED: THE NEXT-NEAREST NEIGHBOR HOPPING TERM

The totla Hamiltonian can be separated into two terms: The NN (Nearest neighbor) one and the NNN (Next-Nearest Neighbor) one

$$\mathcal{H}' = \mathcal{H}_{\text{NN}} + \mathcal{H}_{\text{NNN}}, \quad (4)$$

with the two terms in second quantization form

$$\mathcal{H}_{\text{NN}} =, \quad (5)$$

$$\mathcal{H}_{\text{NNN}} = . \quad (6)$$

Hamiltonian matrix in momentum space

$$\tilde{\mathcal{H}}' = - \begin{pmatrix} \epsilon_A & t_1(1 + e^{ik_1}) & t_2(1 + e^{ik_2}) \\ t_1(1 + e^{-ik_1}) & \epsilon_B & t_3(1 + e^{-ik_3}) \\ t_2(1 + e^{ik_2}) & t_3(1 + e^{ik_3}) & \epsilon_C \end{pmatrix}. \quad (7)$$

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- [1] Wikipedia, Trihexagonal tiling, [https://en.wikipedia.org/wiki/Trihexagonal\\_tiling#Kagome\\_lattice](https://en.wikipedia.org/wiki/Trihexagonal_tiling#Kagome_lattice).